Artificial Intelligence

Course Code: CS-202

Topic: Intelligent Agents and their types, Rationality

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Today’s Goal

- Rationality and its types
- Properties of Task Environment
- Types of Agents

The Concept of Rationality

Rationality

- Rationality means making the best possible decision based on the information available. A rational agent chooses actions that lead to the best expected outcome.
- **Factors influencing rationality:**
  - Performance Measure – How success is defined for the agent.
  - Percept Sequence – What the agent has sensed or observed over time.
  - Knowledge of the Environment – What the agent knows about the world.
  - Available Actions – The choices the agent can make.

Types of Rationality

- Perfect Rationality
- Bounded Rationality
- Instrumental Rationality
- Epistemic Rationality
- Practical Rationality

Types of Rationality

Perfect Rationality

- The agent always makes the best possible decision based on complete knowledge and unlimited computational resources.
- **Example:** A chess AI that calculates every possible move to guarantee the best outcome.

Bounded Rationality

- Real-world agents have limitations in computational resources, knowledge, and time, leading them to make decisions that are "good enough.“
- **Example:** A human making a quick decision with incomplete information.

Types of Rationality

Instrumental Rationality

- The agent selects actions that best achieve its goals based on the given performance measure.
- **Example:** A self-driving car choosing the safest and fastest route to a destination.

Epistemic Rationality

- The agent forms beliefs based on logic, evidence, and reasoning to improve decision-making.
- **Example:** A spam filter updating its rules based on new email patterns.

Types of Rationality

Practical Rationality

- The agent balances goals, risks, and available resources to make reasonable decisions in real-world situations.
- **Example:** A business deciding between investing in marketing or product development based on expected returns.

Environment Characteristics

They describe how the environment behaves and how it affects the agent's decision-making.

- Fully observable (vs. partially observable)
- Deterministic (vs. stochastic)
- Episodic (vs. sequential)
- Static (vs. dynamic)
- Discrete (vs. continuous)
- Single agent (vs. multiagent)

Properties of Task Environments

Fully Observable vs. Partially Observable:

- Fully Observable: The agent has access to complete information. E.g. Chess game
- Partially Observable: Some aspects of the environment are hidden. E.g. Poker game

Deterministic vs. Stochastic

- Deterministic: Next state is completely determined by current state and action. E.g. Solving a maze
- Stochastic: Multiple outcomes are possible for the same action. E.g. Rolling a dice in a board game

Episodic vs. Sequential

- Episodic: The agent's experience is divided into independent episodes.
- Sequential: Actions have long-term consequences.

More Properties of Task Environments

Static vs. Dynamic:

- Static: The environment doesn’t change while the agent is thinking.
- Dynamic: The environment changes over time.

Discrete vs. Continuous:

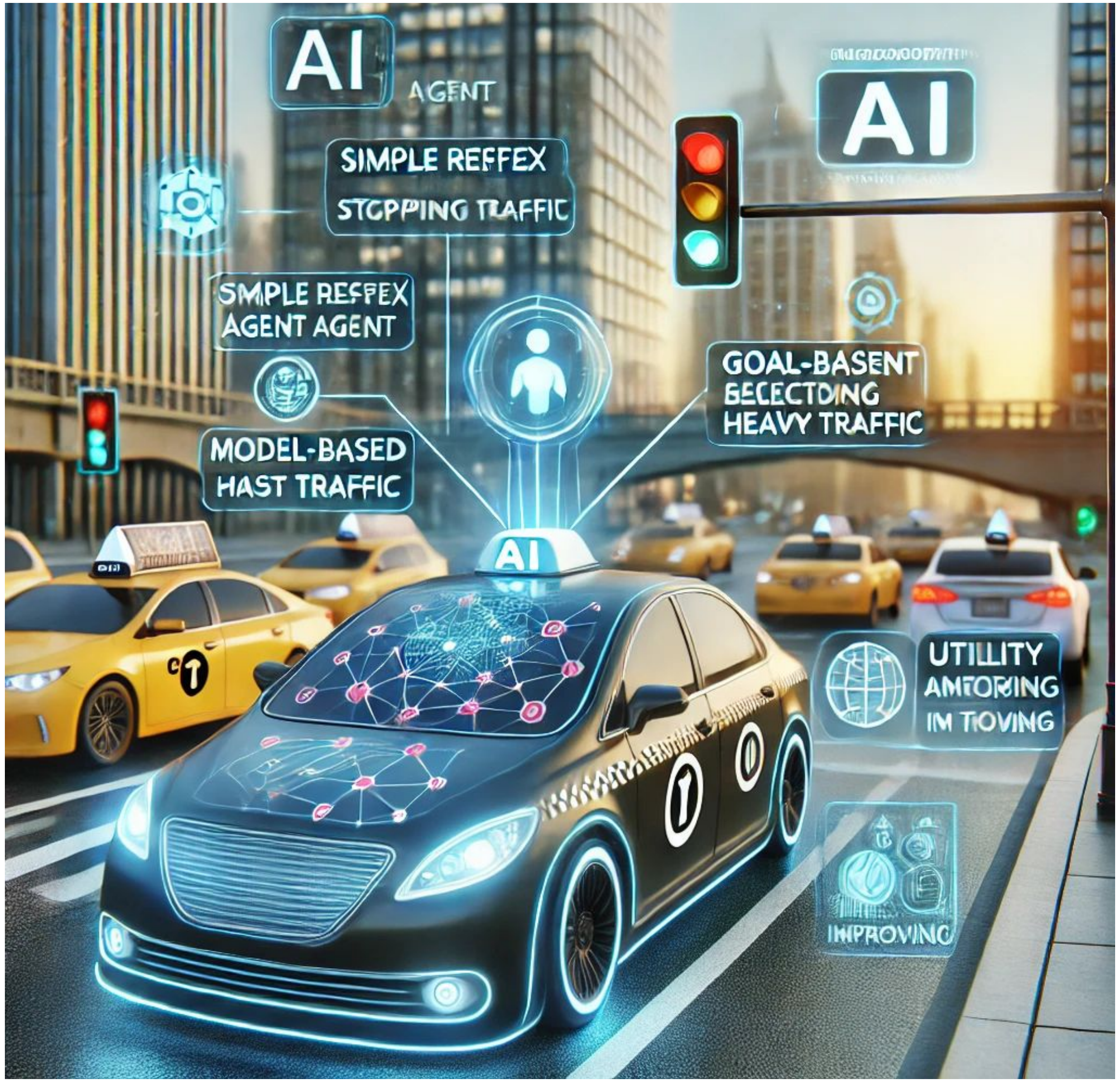
- Discrete: Limited set of actions and states.
- Continuous: Infinite number of possible actions and states.

Single-Agent vs. Multi-Agent:

- Single-Agent: One agent interacts with the environment.
- Multi-Agent: Multiple agents interact with each other.

Example 1: Taxi Driver

- **Discrete or continuous**
    - Continuous
- **Fully Observable or partially observable**
    - partially
- **Single agent or multi-agent**
    - Multi-agent
- **Static or dynamic**
    - dynamic
- **Deterministic or stochastic**
    - Stochastic
- **Episodic or sequential**
    - Sequential



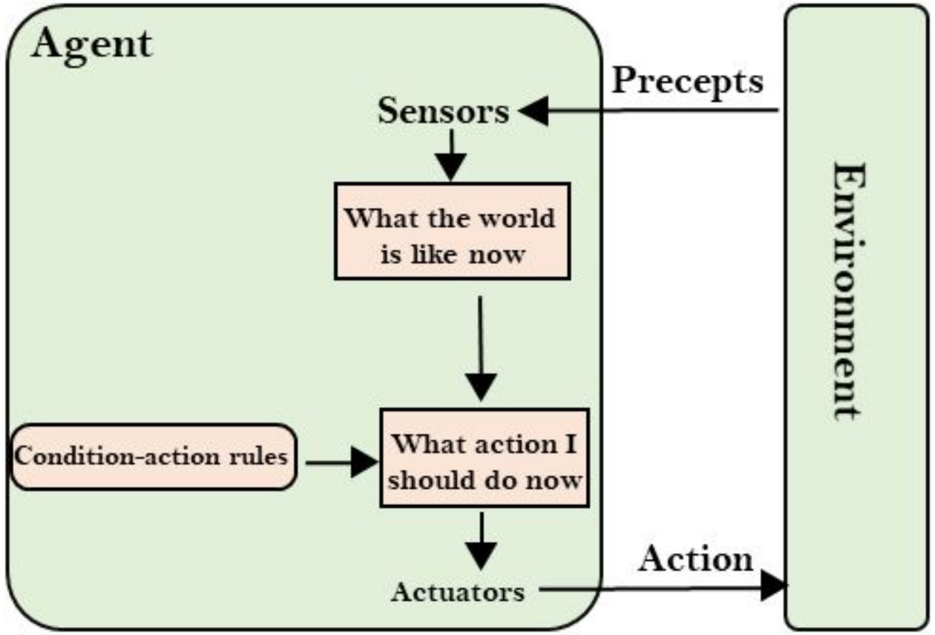
Example 2: Chess with a clock

- **Discrete or continuous**
  - Discrete
- **Fully Observable or partially observable**
  - Fully
- **Single agent or multi-agent**
  - Multi-agent
- **Static or dynamic**
  - Semi-static
- **Deterministic or stochastic**
  - Deterministic (strategic)
- **Episodic or sequential**
  - Sequential

Types of Agents

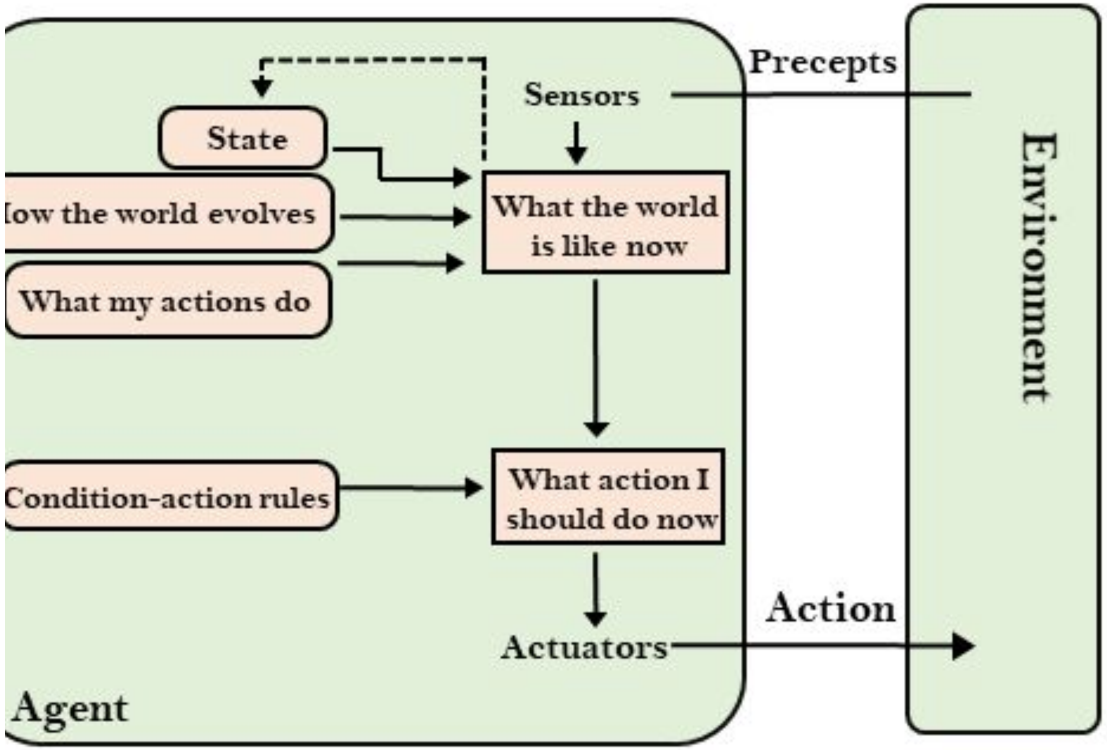
Simple Reflex Agents

- These agents act based on the current perception without considering history.
- They follow a set of predefined rules (condition-action rules).
- Example: A thermostat – it turns the heater on if the temperature is below a set threshold and turns it off otherwise.



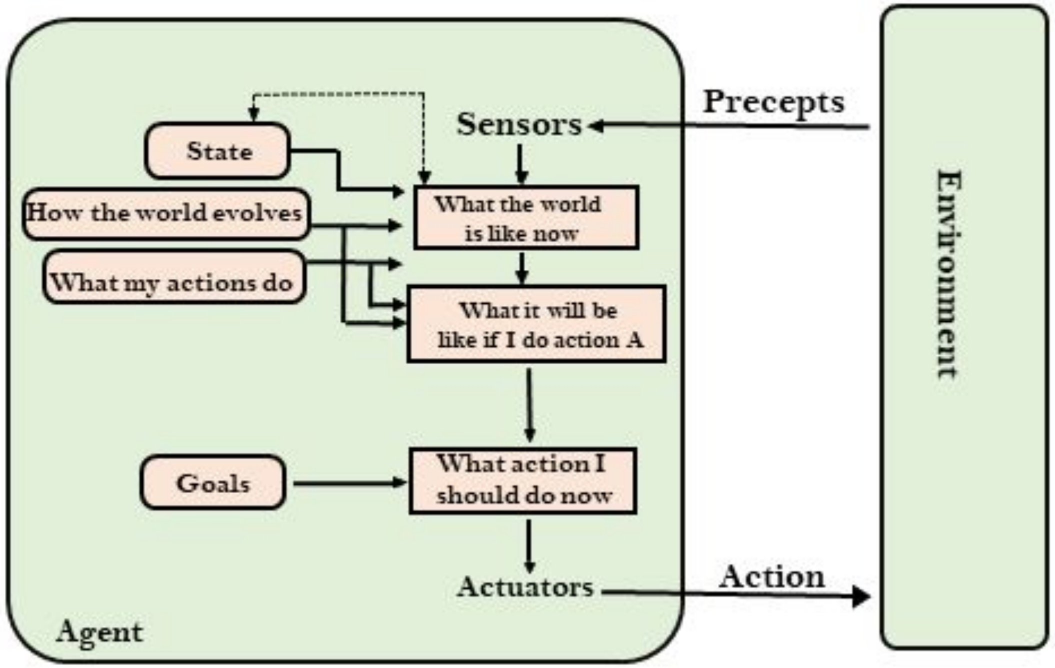
Model-Based Reflex Agents

- These agents maintain an internal model of the environment to handle partially observable environments.
- They use past perceptions to make better decisions.
- Example: A self-driving car at a blind intersection – it remembers that cars may be coming even if they are temporarily out of sight.



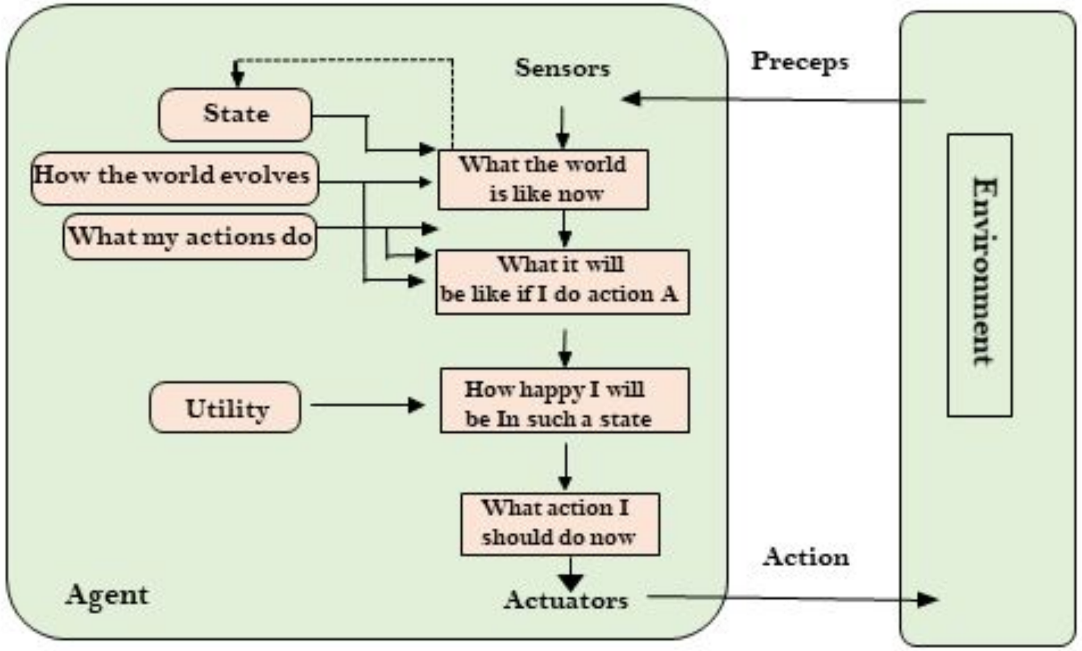
Goal-Based Agents

- These agents make decisions based on goals rather than just rules.
- They evaluate different possible actions to choose the one that helps them achieve their goal.
- Example: A path-finding robot – it plans the shortest route to reach a destination instead of just reacting to obstacles.



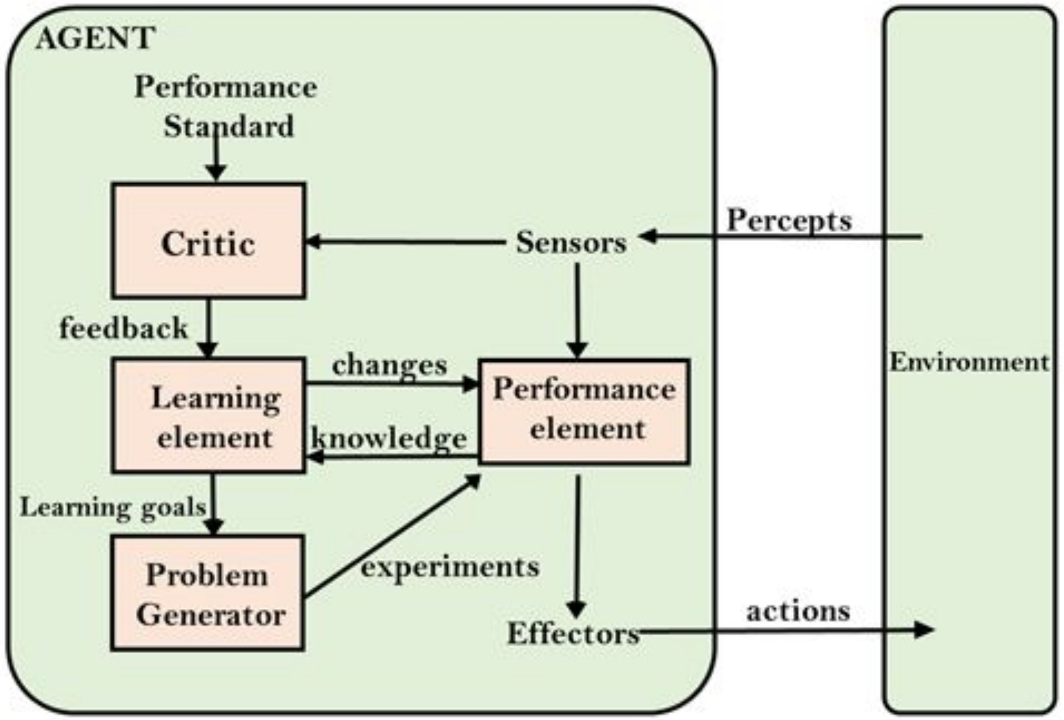
Utility-Based Agents

- These agents consider not just achieving a goal but also optimizing performance using a utility function (which assigns values to different states).
- Example: A shopping recommendation system – it suggests products based on how probable a user is to buy and be satisfied, rather than just showing random items.



Learning Agents

- Learning agents improve their performance over time by gaining experience, updating their knowledge, and adjusting their behavior.
- Example: A spam filter – it learns from user feedback and improves at detecting spam emails over time.



Summary

- Rationality is goal-oriented behaviour based on information available expected performance.
- Task environments vary in observability, determinism, dynamics, and agent interaction.
- Agent structures range from simple reflex agents to more complex utility-based agents.

Practice Task

Identify how these agents interact with their environment based on the environment properties.

- Robot Vacuum Cleaner (e.g., Roomba)
- Chatbot (e.g., AI Customer Service Assistant)

Any Question?